

DSA Test

First

```
import java.util.*;
import static java.lang.System.out;

public class First {
    public static void main(String[] args) {
        Scanner sc= new Scanner(System.in);
        int n=sc.nextInt();
        System.out.print("Choose 1 for new way and 2 for old way");
        int ch=sc.nextInt();

        if(ch==1)
            newway(n);

        if(ch==2)
            oldway(n);
    }
    public static void newway (int n) {
        if(n%3==0) {
            out.print("fun");
        }
        if(n%7==0) {
            out.print("buzz");
        }
        out.println();
        System.out.print("newway");
    }
    public static void oldway (int n) {
        if(n%3==0 && n%7==0) {
            out.print("fun-buzz");
        } else if(n%3==0) {
            out.print("fun");
        } else if(n%7==0) {
            out.print("buzz");
        } else {
            out.print(n);
        }
        out.println();
        System.out.print("oldway");
    }
}
```

Second

```
import java.util.*;
public class Second {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number");
        int a = sc.nextInt();
        System.out.print("Enter last number");
        int b = sc.nextInt();
        System.out.print("Choose 1 for new way and 2 for old way");
        int ch=sc.nextInt();
        if(ch==1)
```

```

        oddnumbers(a, b);

        if(ch==2)
            oddnumbersoldway(a, b);

    }
    static void oddnumbers(int a, int b) {
        boolean s=isodd(a);
        if(s==false)
            a=a+1;
        for(int i=a;i<=b;i=i+2){
            System.out.print(i+" ");
        }
    }
    public static boolean isodd(int a) {
        return a%2!=0;
    }
    static void oddnumbersoldway(int a, int b) {
        System.out.print("oldway");
        for(int i=a;i<=b;i++) {
            if(i%2!=0) {
                System.out.print(i+" ");
            }
        }
    }
}

```

Third

```

import static java.lang.System.out;

import java.util.Scanner;
public class Third {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        ispalindrome(n);
    }
    static void ispalindrome(int n) {
        int rev = 0;
        int original=n;
        while(n>0) {
            int digit = n % 10;
            rev = rev * 10 + digit;
            n = n / 10;
        }
        System.out.print("ans is "+rev+" ");
        if(rev == original) {
            System.out.print(" is palindrome");
        } else {
            System.out.print("not palindrome");
        }
    }
}

```

Fourth

```
import java.util.Scanner;
public class Fourth {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        fibbo(n);
    }

    static void fibbo(int n) {
        int a=0,b=1;
        if(n<=0){
            System.out.print("Invalid input");
            return;
        }

        if (n==1){
            System.out.print(a+" ");
            return;
        } else if(n==2){
            System.out.print(a+" "+b+" ");
            return;
        }
        System.out.print(a+" "+b+" ");
        if(n>2){
            for(int i=2;i<n;i++) {
                int c=a+b;
                System.out.print(c+" ");
                a=b;
                b=c;
            }
        }
    }
}
```